BUSINESS INTELLIGENCE · IIB423T-1 · C1 PROJECT

How much does the start decide a Formula 1 race?

Exploratory data analysis of the 2014–2024 World Championship

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01 · THE PROJECT

One question, one user, one decision

How strongly does the starting position decide the final position in 2014–2024, does it vary by circuit, and how much of it is really the car?

Parc fermé: the car that qualifies is the car that races, so a team has one setup for both sessions (FIA sporting regulations).

PROPOSED USER

A team strategist (our proposal)

DECISION

How far to lean the single setup toward qualifying pace or race pace, circuit by circuit

WHY THIS LINE, NOT PIT STOPS (W1)

Grid and result are observed directly; pit-stop data only record total pit-lane time, not the crew's work

02 · THE KPI

IDC: how much the grid decides the finishing order

MAIN KPI · ONE VALUE PER CIRCUIT

IDC · Grid Determinism Index

```
ρ_race = Spearman(grid, final)
IDC = median(ρ_race) + 95 % CI
```

Finishers only · interval from 5,000 bootstrap resamples of the circuit's races

IDC near 1 → you finish where you start → lean the setup toward qualifying

SUPPORT

P(top 10 | grid block)

What each grid slot is worth in points

SUPPORT

Pole conversion

Share of races won by the driver starting P1

CONTROL

Partial ρ

Grid → result after removing the team's strength

Ordinal positions → Spearman

Skewed, capped at 1 → median

Few races per circuit → interval

03 · THE PROCESS

Three passes over the data, each with its own question

W1 · DONE

Is it feasible?

Proxy inspection of two candidate lines

→

NOTEBOOK 01

What do we have?

EDA of the raw dataset: 12 CSV files, 1950–2024

→

NOTEBOOK 02

What can we trust?

Cleaning → analytic table, 4,606 rows × 42 columns

→

NOTEBOOK 03

What does it tell us?

EDA driven by the KPI → 5 KPI tables

WHERE EACH STEP OF THE ASSIGNMENT LIVES

1 · Idea & KPI → slides 2–3

2 · GitHub repository

3 · Download → data/raw

4 · Explore → notebook 01

5 · Clean → notebook 02

6 · Full EDA → notebook 03

7 · Upload to GitHub

EDA 1 · WHAT DO WE HAVE?

12 related tables, and every join checks out

Table	One row =	Years
results	one driver in one race	1950–2024
qualifying	one driver's qualifying	1994–2024
lap_times	one lap of one driver	1996–2024
pit_stops	one pit stop	2011–2024
constructor_standings	one team after a race	1958–2024
sprint_results	one driver in a sprint	2021–2024
races + 4 lookups	one entity per row	—

Keys: 11 of 12 unique

results repeats 91 (race, driver) keys: shared cars, 1950–1978

Integrity: 0 orphans

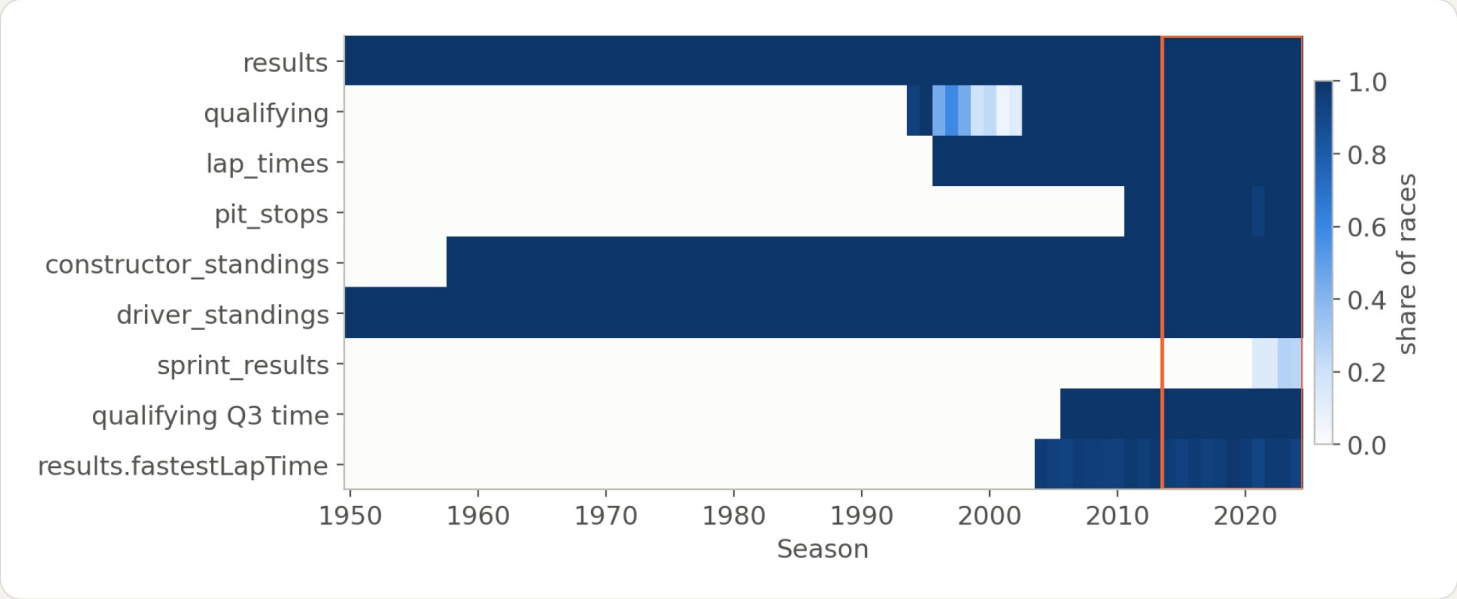
14 relationships tested, e.g. results.driverId → drivers

⚠ Special codes

\N = missing · grid 0 = did not qualify (old eras) or pit-lane start (modern)

EDA 1 · WHAT DO WE HAVE?

Why 2014–2024: complete tables, comparable seasons



Share of each season's races with data, by table. Orange frame = 2014–2024.

COVERAGE

Qualifying complete from 2003, Q3 from 2006, pit stops from 2011

POINTS

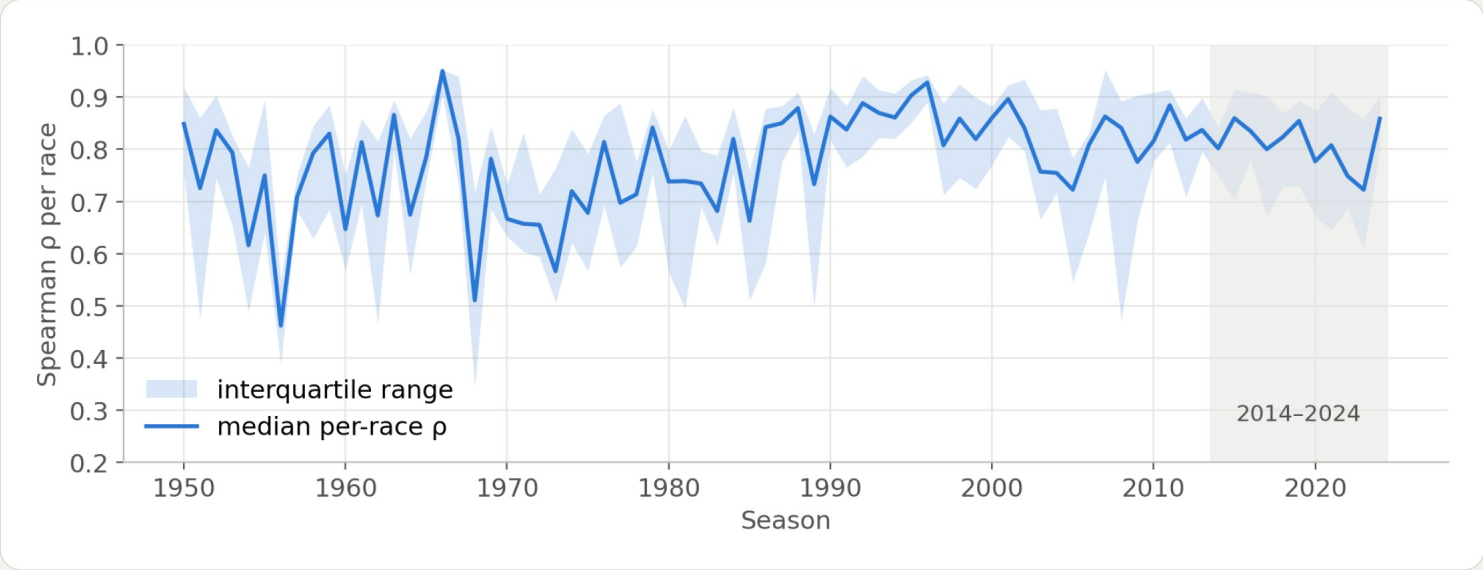
8–10 to the winner for decades, 25 since 2010 → we use positions

RELIABILITY

Under 50 % of cars finished before the 1990s; about 80–90 % today

EDA 1 · WHAT DO WE HAVE?

A first look: the grid has always mattered



Median Spearman ρ(grid, final) per season, finishers only; band = interquartile range.

0.75–0.9

median ρ per race since the late 1980s

0.77

final position ↔ grid, 2004–2024

0.64

grid ↔ fastest-lap rank: the car drives both

CLEANING · WHAT CAN WE TRUST?

From 26,759 raw results to one analytic table

26,759
raw results, 1950–2024

→

4,626
results in 2014–2024

→

4,606
without Belgium 2021

→

4,606 × 42
analytic table · 227 races

Decision	Why
Exclude Belgium 2021	1 lap behind the safety car: order = grid, so ρ = 1 by construction
grid 0 → pit-lane flag, grid = NaN	a pit-lane start is not a position smaller than 1
Qualifying time → seconds → % gap	comparable across circuits of different lengths
Race pace = median green lap	no lap 1, no pit in/out laps, no laps over 107 % of the race median
Team rank after the previous round	standings after the same race contain the result: leakage
20 team ids → 12 lineages	Force India → Racing Point → Aston Martin is one team

CLEANING · WHAT CAN WE TRUST?

Every join and rule is tested, not assumed

```
df = df.merge(qualifying, on=["raceId", "driverId"],
              how="left", validate="one_to_one",
              indicator=True)
# both: 4,590 · left_only: 16 · right_only: 0
assert len(df) == n0  # no row lost or duplicated
```

9 of 9 validation checks pass

- unique (race, driver) key
- one winner per race
- final positions 1..n
- pit-lane starts flagged
- gaps to fastest ≥ 0
- Belgium 2021 excluded

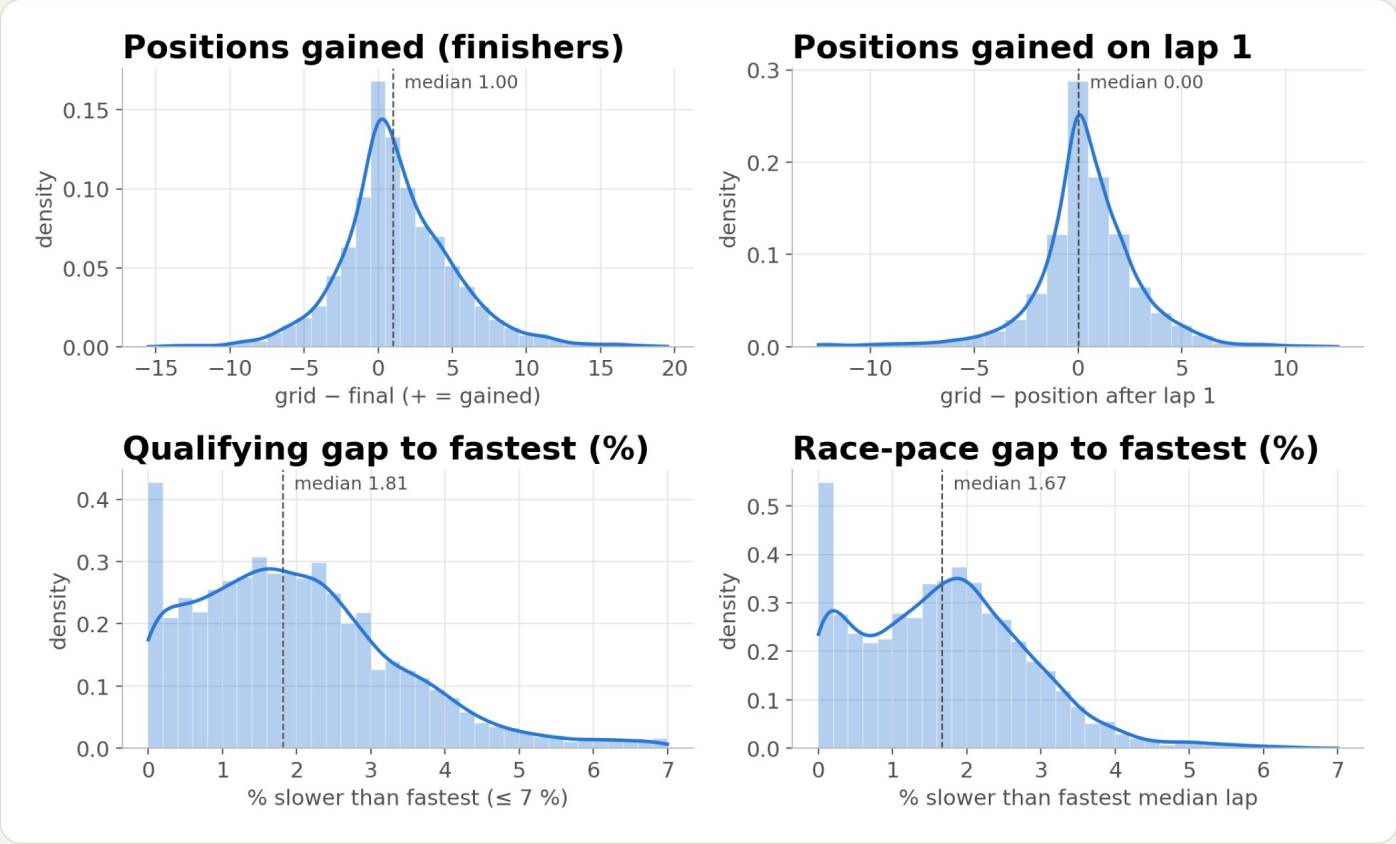
Missing values have a reason

Column	Missing	Why
position	15.7 %	not classified
race pace	11.3 %	retired too early
lap-1 position	3.0 %	did not finish lap 1
grid	1.7 %	pit-lane start
team rank	0.1 %	team's first race

→ **No imputation: each gap has a meaning**

EDA 2 · WHAT DOES IT TELL US?

Distributions first: skewed data, robust methods



Lap 1: peak at 0 and a long left tail; the big moves are losses

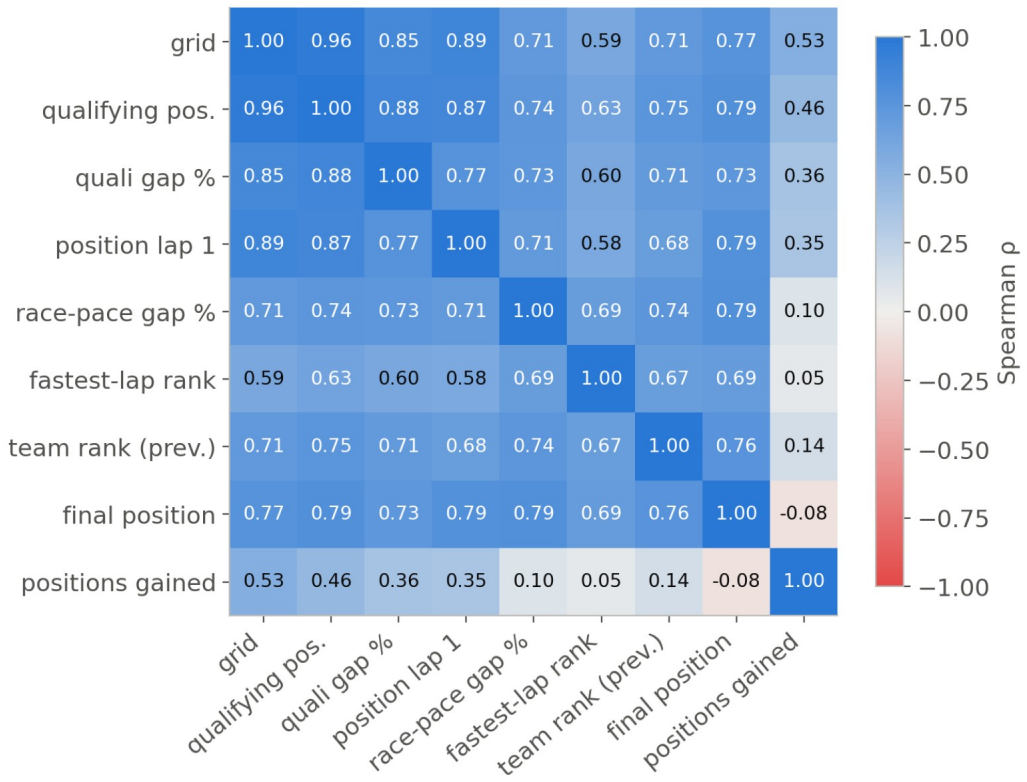
Qualifying and race-pace gaps: right-skewed, most cars within 1–3 % of the fastest

Per-race ρ: median 0.81, skew −1.7, Shapiro-Wilk $p < 0.001$

So: medians, bootstrap intervals and rank-based tests

EDA 2 · WHAT DOES IT TELL US?

Correlations: everything "fast" moves together



0.76–0.79

final position vs race pace, qualifying, lap 1, grid and team rank

0.71

grid vs team rank → control for the car

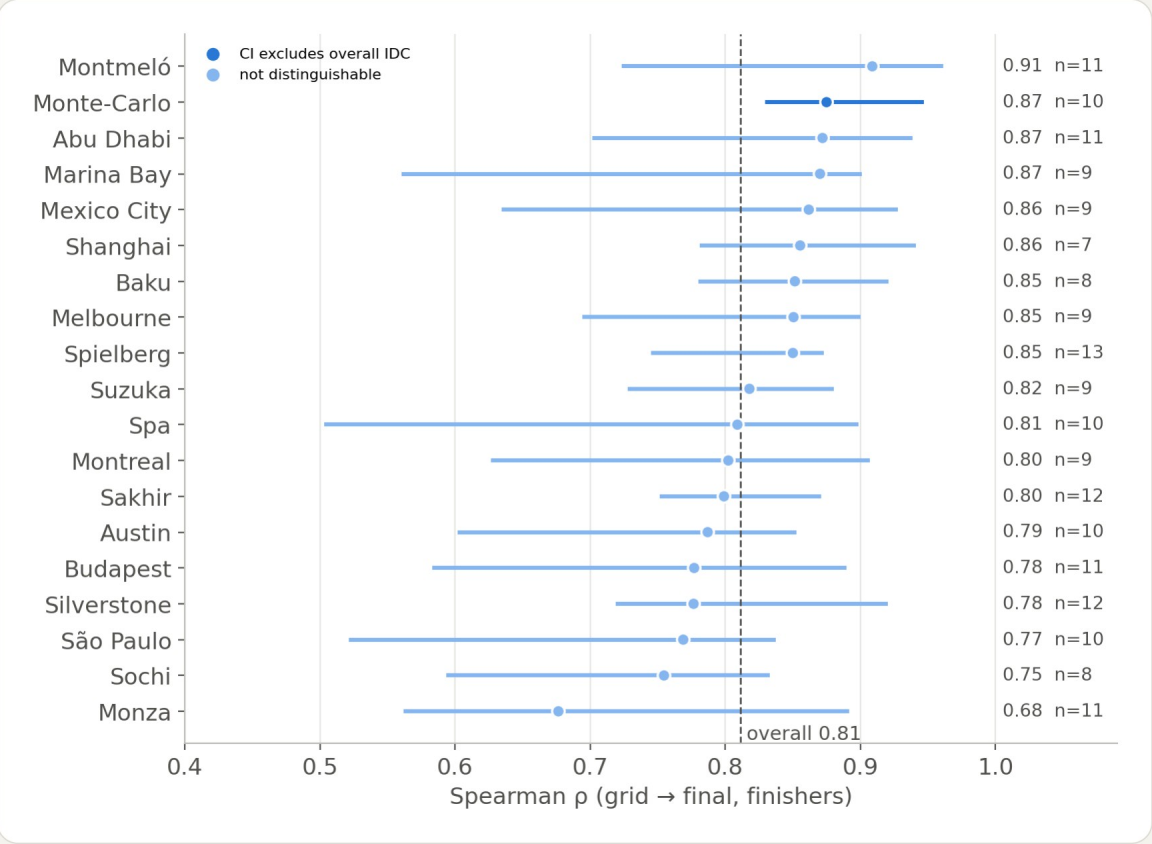
0.53

positions gained vs grid (room to gain)

Multicollinearity: no single variable wins on its own, a warning for the ML stage.

MAIN KPI · IDC BY CIRCUIT

Strong everywhere, different almost nowhere



0.81

overall IDC · 95 % CI 0.79–0.83

0.68 → 0.91

Monza → Montmeló: intervals overlap

1 of 19

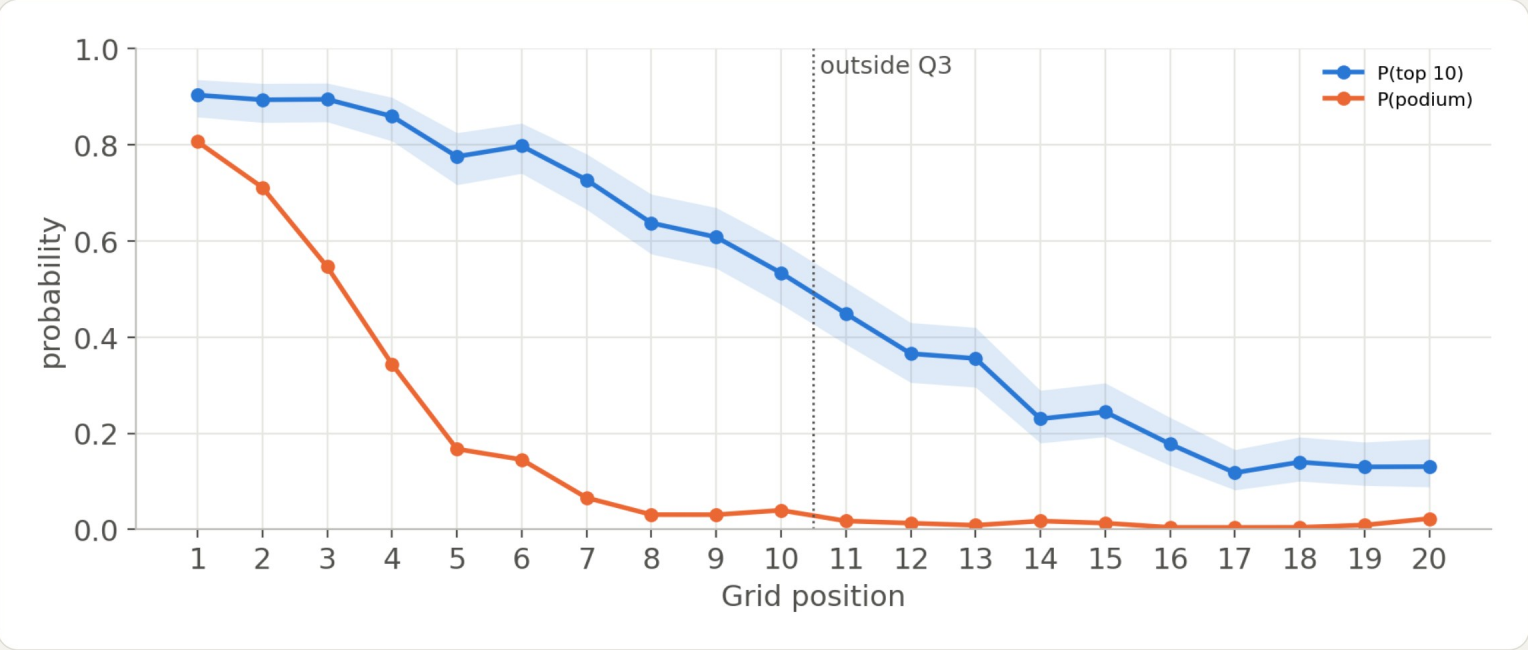
clearly above: Monaco (Monte-Carlo)

p = 0.29

Kruskal-Wallis: no significant difference

SUPPORT KPIS · P(TOP 10 | GRID) AND POLE CONVERSION

What a grid slot is worth



Probability of finishing in the top 10 (blue) or on the podium (orange) by grid position, all starters, 2014–2024; band = 95 % Wilson interval.

90 %
score points from P1–P3

33 %
score points from P11–P15

-7 pts
per slot lost, on average, from P7 to P14

52 %
of poles become wins

CONTROL KPI · PARTIAL P

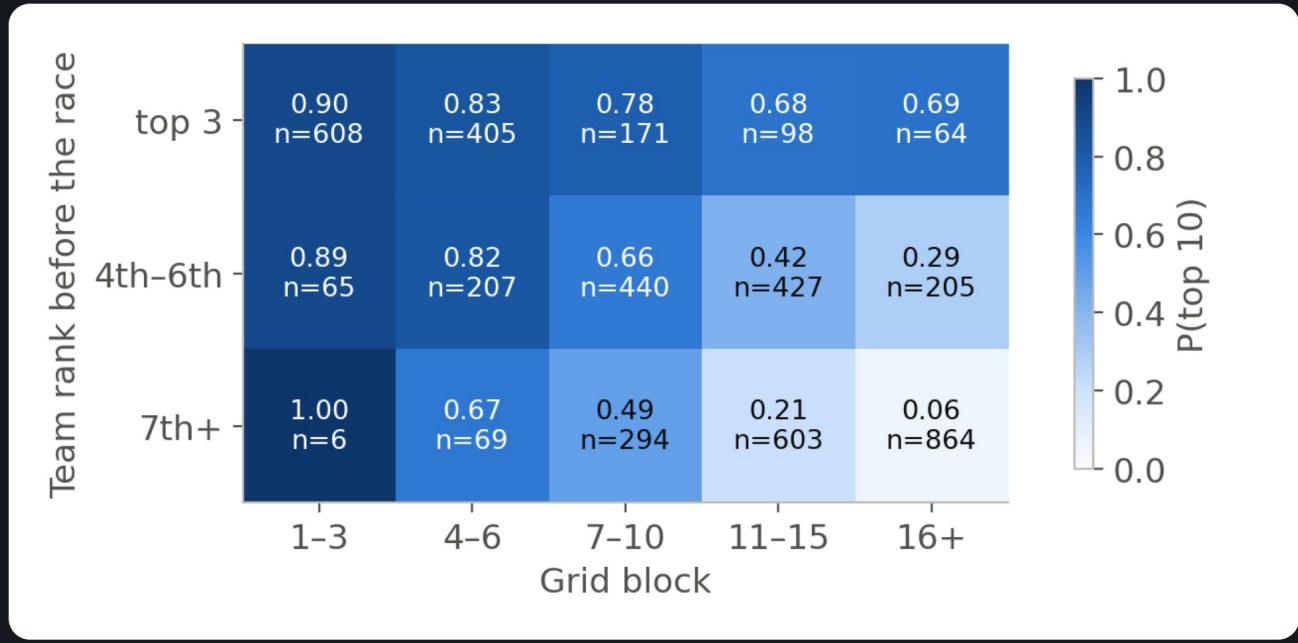
Is it the start, or the car?

0.77 → **0.50**

raw ρ(grid, final)

team strength removed · CI 0.46–0.53

Shared variance falls from 59 % to 25 %.
Track position still matters on its own.



A fast car recovers from a bad grid slot; a slow one does not.

Top-3 team from P11–P15: 68 % score · 7th+ team from P16+: 6 %

CONCLUSIONS

What we tell the strategist

1 The start matters, consistently

IDC 0.81 overall; 40 % of position changes happen on lap 1

2 A large part of it is the car

ρ falls from 0.77 to 0.50 once team strength is removed

3 Circuits barely differ

Only Monaco stands out; Kruskal-Wallis p = 0.29

4 Qualifying pace is the lever

The qualifying order predicts the result better than the grid (0.84 vs 0.81)

Robustness: the same answer without sprint weekends, small circuits or the 7 most chaotic races (IDC 0.81–0.82).

LIMITATIONS

No weather, safety-car or tyre data · team strength is a coarse proxy · only 7–13 races per circuit

NEXT STEPS

ML: predict a top-10 finish against the baseline "you finish where you start" · Dashboard: one card per circuit

Thank you. Questions?

Everything is reproducible: open GROUP_06_C1 and run notebooks 01 → 02 → 03

Repository: github.com/BeTRoCo/GROUP_06_C1

Data: Formula 1 World Championship (1950–2024), Kaggle (rohanrao), from the Ergast API · downloaded 10 September 2026